MECHANICAL LEGEND	
SYMBOL	DESCRIPTION
	DUCT WIDTH x DEPTH CLEAR INSIDE DIMENSION
	SUPPLY AIR DUCT SECTION UP - DOWN
	RETURN OR EXHAUST AIR DUCT SECTION UP - DOWN
	ROUND SUPPLY AIR DUCT SECTION UP - DOWN
	ROUND RETURN AIR DUCT SECTION UP - DOWN
V.D.	MANUAL VOLUME DAMPER
	SQUARE ELBOW W/TURNING VANES
DN	DROP IN DUCT (DIRECTION OF FLOW)
UP	RISE IN DUCT (DIRECTION OF FLOW)
	TAKE-OFF DAMPER/EXTRACTOR
	NEW CONSTRUCTION
G	GAS PIPE
	CONDENSATE DRAIN PIPE
	REFRIGERANT PIPE
	REDUCER
	UNION
	PIPE DOWN
	PIPE UP
	GAS PRESSURE REDUCING VALVE
	ISOLATION VALVE
	RECTANGULAR TO ROUND TRANSITION
	SUPPLY DIFFUSER
	SUPPLY GRILL
	EXHAUST GRILL
	RETURN GRILL
EXH	EXHAUST AIR
GR.	GRILLE
PRV	PRESSURE REDUCING VALVE
EF	EXHAUST FAN
V.D.	VOLUME DAMPER
V.I.F.	VERIFY IN FIELD
W.C.	WATER COLUMN
T	THERMOSTAT
A.F.F.	ABOVE FINISHED FLOOR
B.O.D.	BOTTOM OF DUCT
BTUH	BRITISH THERMAL UNITS PER HOUR
CD	CONDENSATE DRAIN
C.O.	CLEAN OUT
CFH	CUBIC FEET PER HOUR
CFM	CUBIC FEET PER MINUTE
CLG.	COOLING
EL.	ELEVATION
FPM	FEET PER MINUTE
HP	HORSEPOWER
HTG.	HEATING
KW	KILOWATT
N.R.	NOT REQUIRED
N.T.S.	NOT TO SCALE
O.A.	OUTSIDE AIR
PSI	POUNDS PER SQUARE INCH
SQ. FT.	SQUARE FEET
AHU	AIR HANDLING UNIT
CEF	CEILING EXHAUST FAN
KEF	KITCHEN EXHAUST FAN

GENERAL NOTES

ALL WORK PERFORMED SHALL CONFORM WITH LOCAL CITY & STATE REGULATIONS. ALL WORK SHALL BE CONDUCTED, INSTALLED AND COMPLETED IN A WORKMANLIKE AND APPROVED MANNER SO AS TO SECURE THE RESULTS INTENDED BY THESE DOCUMENTS

ALL WORK IS TO BE FULLY COORDINATED WITH ALL OTHER TRADES.

CONTRACTOR IS RESPONSIBLE FOR IDENTIFYING AND INCLUDING ANY ITEMS NOT INDICATED ON THE DRAWINGS BUT NECESSARY FOR PROPER OPERATION OF MECHANICAL SYSTEM.

THE SEQUENCE FOR THE INSTALLATION OF ALL WORK SHALL BE COORDINATED BETWEEN ALL CONTRACTORS ON THE PROJECT & IN STRICT ACCORDANCE WITH ARCHITECT/ENGINEER & OWNER'S STIPULATION.

THE CONTRACTOR SHALL COORDINATE ALL WORK WITH OTHER TRADES & SHALL MAKE NECESSARY OFF-SETS & CHANGES IN ELEVATIONS TO ACCOMMODATE OTHER TRADES & THE EXISTING CONDITIONS.

WHERE THERE IS EVIDENCE THAT WORK OF ONE TRADE WILL INTERFERE WITH WORK OF OTHER TRADES, ALL TRADES SHALL MEET ON JOB SITE TO WORK OUT SPACE CONDITIONS & MAKE SATISFACTORY ADJUSTMENTS TO INSTALLATION OF THE NEW WORK. CONTRACTORS SHALL VERIFY EXACT LOCATIONS OF ALL DEVICES & EQUIPMENT WITH FIELD CONDITIONS, SHOP DRAWINGS, & WORK OF OTHER TRADES PRIOR TO ROUGH IN. EACH CONTRACTOR SHALL BE RESPONSIBLE, AT THEIR OWN EXPENSE, FOR THE REMOVAL & REINSTALLATION OF ANY PART OF THEIR WORK IF SAME WAS INSTALLED WITHOUT CONSULTING WITH OTHER TRADES BEFORE INSTALLING THEIR WORK.

ALL EQUIPMENT AND MATERIALS SHALL BE U.L. LISTED.

CONTRACTOR SHALL SECURE AND PAY FOR ALL PERMITS AND FEES REQUIRED FOR THEIR WORK.

CONTRACTOR SHALL REFER TO THE ARCHITECTURAL & STRUCTURAL CONTRACT DRAWINGS (BEFORE SUBMITTING THEIR BIDS) TO FAMILIARIZE THEMSELVES WITH THE EXTENT OF THE GENERAL CONTRACTORS WORK, CEILING HEIGHTS AND CLEARANCE FOR INSTALLING THEIR WORK

INCLUDE ALL OVERTIME NECESSARY TO MAINTAIN JOB SCHEDULE UNDER NORMAL CONDITIONS OR DUE TO THIS CONTRACTOR'S NEGLIGENCE OR INABILITY TO PROPERLY STAFF THE PROJECT.

CONTRACTORS MUST KEEP UP WITH THE PROGRESS OF THE JOB.

ALL MAJOR PIECES OF MECHANICAL EQUIPMENT SHALL BE STARTED AND ADJUSTED AND PUT INTO OPERATION BY A FACTORY REPRESENTATIVE. ALL EQUIPMENT SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURERS INSTALLATION INSTRUCTIONS.

PRIOR TO BID, IF THE CONTRACTOR FINDS ANY DISCREPANCIES OR OMISSIONS IN THE PROJECT DOCUMENTS, THE CONTRACTOR IS TO NOTIFY THE ENGINEER IN WRITING & OBTAIN CLARIFICATION. ADDITIONAL COMPENSATION WILL NOT BE GRANTED AFTER AWARD OF CONTRACT FOR ANY ADDITIONAL WORK REQUIRED TO COMPLY WITH THESE DOCUMENTS.

ALL CHANGE PROPOSAL REQUESTS FOR WORK ADDITIONAL TO THE BASE BID CONTRACT SHALL BE BASED ON MATERIAL, LABOR, OVERHEAD AND PROFIT AS PUBLISHED IN THE LATEST EDITION OF "MEANS MECHANICAL, ELECTRICAL, PLUMBING AND BUILDING CONSTRUCTION COST DATA. "ALL CHANGE REQUESTS MUST BE BROKEN DOWN IN THE FOLLOWING MANOR.
MATERIAL COST: (IE. EQUIPMENT, SHEET METAL PER POUND AND PIPING PER LINEAL FOOT/FITTING)
LABOR COST: (NUMBER OF HOURS AT CURRENT LABOR RATE PER HOUR)
OVERHEAD & PROFIT: (INDICATING PERCENTAGES)
TOTAL CHANGE ORDER PRICE: (MATERIAL + LABOR + O&P)
PRICING FOR ALL ITEMS OF WORK WHICH ARE TO BE CREDITED TO THE PROJECT SHALL BE BROKEN DOWN IN A SIMILAR MANOR TO THE ADDED COSTS.

THE ASSOCIATED COST FOR DRAFTING CHANGES (INCLUDING THREE DIMENSIONAL MODELING) SHALL NOT EXCEED 10% OF THE COST OF MATERIAL AND LABOR FOR THE CHANGE

ALL CUTTING AND PATCHING THAT IS REQUIRED TO COMPLETE THE WORK SHALL BE THE RESPONSIBILITY OF THE INSTALLING CONTRACTOR.

THE CONTRACTOR IS TO PROVIDE ALL LINTELS, SUPPORT STEEL AND FRAMING THAT IS REQUIRED TO COMPLETE THE WORK.

CONTRACTOR SHALL PROVIDE SLEEVES IN BEAMS, FLOORS, AND COLUMNS AND WALLS AS SHOWN ON DRAWINGS, AS REQUIRED BY JOB SITE CONDITIONS, AND/OR SPECIFIED, WHEN INSTALLING THEIR WORK. ALL BEAMS AND COLUMNS WHICH ARE REQUIRED TO BE SLEEVED SHALL BE CUT AND REINFORCED AS REQUIRED BY FIELD CONDITIONS AND LOCATIONS AND SIZES SHALL BE CHECKED AND APPROVED BY ARCHITECTS BEFORE CONTRACTOR CUTS ANY STRUCTURAL BUILDING MEMBER.

PROVIDE ALL COORDINATION AND MISCELLANEOUS IRON NECESSARY FOR SUITABLE ANCHORAGE OF HVAC ITEMS AND EQUIPMENT.

CONTRACTOR IS TO INCLUDE ALL REQUIRED PREMIUM TIME IN BASE BID.
INCLUDE ALL ESCALATION COSTS REQUIRED TO COMPLETE THE WORK.

PROVIDE FOR SAFETY AND PROTECTION OF CONTRACTOR'S OWN WORK, INCLUDING THE COVERING OF ANY HOLES, SHAFT OPENINGS, ETC., SO AS TO AVOID ANY UNNECESSARY SAFETY HAZARDS AS REQUIRED AND OUTLINED BY OSHA AND ALL APPLICABLE REGULATIONS.

PROVIDE DUST AND NOISE PROTECTION OF ADJOINING NON-CONSTRUCTION AREAS. PROPERLY PROTECT ALL FLOORS, ROOFS AND THE LIKE.

MECHANICAL EQUIPMENT & APPLIANCES SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR THE LABELED EQUIPMENT. CONNECTIONS TO THE MECHANICAL EQUIPMENT AND APPLIANCES, SUCH AS FUEL SUPPLY, CHIMNEY & DUCTS, SHALL CONFORM TO THE REQUIREMENTS OF THESE DOCUMENTS. MANUFACTURER'S INSTALLATION INSTRUCTIONS SHALL BE AVAILABLE ON THE JOBSITE AT ALL TIMES FOR INSPECTION.

THE DRAWINGS, SCHEDULES, & SPECIFICATIONS HAVE BEEN PREPARED USING ONE MANUFACTURER FOR EACH TYPES OF EQUIPMENT AS THE BASIS FOR DIMENSIONAL & MECHANICAL DESIGN. SUBSTITUTIONS FOR PRODUCTS WILL ONLY BE CONSIDERED IF SUBMITTED 10 DAYS PRIOR TO BID DATE & ONLY FOR PRODUCTS EQUAL OR BETTER THAN THAT SPECIFIED. THE CONTRACTOR SHALL BE RESPONSIBLE FOR CHECKING ALL THE DIMENSIONS OF THE EQUIPMENT TO VERIFY THAT IT WILL FIT IN THE SPACE SHOWN ON THE DRAWINGS. MINOR DEVIATIONS IN DIMENSIONS WILL BE PERMITTED; PROVIDED THE RATINGS MEET THOSE SHOWN ON THE DRAWINGS AND EQUIPMENT WILL PHYSICALLY FIT INTO THE SPACE ALLOCATED WITH SUITABLE ACCESS AROUND EQUIPMENT FOR OPERATION & MAINTENANCE ON THE EQUIPMENT.

THE MECHANICAL EQUIPMENT HAS BEEN COORDINATED WITH THE ELECTRICAL DESIGN DRAWINGS BASED ON THE ELECTRICAL CHARACTERISTICS OF THE EQUIPMENT SPECIFIED. ALL CHANGES AND/OR MODIFICATIONS TO THE ELECTRICAL DESIGN AND INSTALLATION EXPENSE, DUE TO SUBSTITUTIONS OF EQUIPMENT (I.E. AMPERAGE INCREASE) WILL BE THE RESPONSIBILITY OF THE MECHANICAL CONTRACTOR.

CONTRACTOR SHALL SUBMIT DETAILED DIMENSIONED SHOP DRAWINGS FOR ALL WORK WHICH MUST BE REVIEWED, COORDINATED & SIGNED OFF BY ALL OTHER TRADES BEFORE SUBMITTAL. IN PREPARATION OF SHOP DRAWINGS, CONTRACTOR MAY, AT HIS OPTION, OBTAIN ELECTRONIC DRAWING FILES IN AUTOCAD FORMAT ON A CD-ROM DISK FROM THE ENGINEER FOR SHIPPING & HANDLING FEE OF $150.00 PER REQUEST. CONTRACTOR SHALL CONTACT THE ARCHITECT & ENGINEER FOR WRITTEN AUTHORIZATION & NECESSARY RELEASE AUTHORIZATION FORM & TO SPECIFY SHIPPING METHOD. IN ADDITION TO PAYMENT, ARCHITECTS WRITTEN AUTHORIZATION & ENGINEERS RELEASE AGREEMENT FORM MUST BE RECEIVED BEFORE ELECTRONIC DRAWING FILES WILL BE SENT.

CONTRACTOR AND/OR MANUFACTURER SHALL VERIFY THAT THE CHARACTERISTICS OF THE EQUIPMENT HE SUBMITS FOR REVIEW MEETS THE CAPACITY AND DUTY SPECIFIED.

WHEN EQUIPMENT IS SUBMITTED FOR REVIEW AND DOES NOT MEET THE PHYSICAL SIZE OR ARRANGEMENT OF THAT SCHEDULED & SPECIFIED, CONTRACTOR SHALL PAY FOR ALL ALTERATIONS REQUIRED TO ACCOMMODATE SUCH EQUIPMENT AT NO ADDITIONAL COST TO OWNER. CONTRACTOR WILL ALSO PAY ALL COSTS FOR ADDITIONAL WORK REQUIRED BY OTHER CONTRACTORS, OWNER, ARCHITECT, OR ENGINEER TO MAKE CHANGE WHICH WOULD ALLOW THE EQUIPMENT TO FIT IN THE SPACE & FUNCTION AS INTENDED.

HVAC NOTES:

- ALL EQUIPMENT, DUCTWORK, PIPEWORK, ETC SHALL BE SUPPORTED.
- ALL DUCT SIZES SHOWN ARE INSIDE CLEAR DIMENSIONS.
- ALL DUCTWORK AND PIPING IS SHOWN SCHEMATICALLY. PROVIDE ALL TRANSITIONS, TURNING VANES, ELBOWS, FITTINGS, ETC. TO ALLOW SMOOTH FLOWS. ALL SPLIT DUCT FITTINGS SHALL TRANSITION TO FULL SIZE OF THE SUM OF BOTH BRANCHES UPSTREAM OF SPLIT.
- MAINTAIN A MINIMUM 6" CLEARANCE BETWEEN DUCTWORK, PIPING, EQUIPMENT, ETC. AND ALL FIRE RATED AND FIRE/SMOKE RATED PARTITIONS, TO ALLOW FOR INSPECTION OF RATED WALLS.
- SLEEVE AND SEAL ALL PIPING PENETRATIONS THROUGH BUILDING PARTITIONS.
- VERIFY AND COORDINATE ALL ROOF, WALL, AND FLOOR PENETRATIONS WITH THE STRUCTURAL AND ARCHITECTURAL DRAWINGS PRIOR TO THE START OF CONSTRUCTION.
- PROVIDE ACCESS PANELS TO ALL CONCEALED VALVES, DAMPERS, AND EQUIPMENT. COORDINATE THE LOCATION OF ACCESS PANELS TO INSURE THAT THE EQUIPMENT CAN BE MAINTAINED ADEQUATELY.
- INSTALL CONDENSATE DRAINS AT A MINIMUM SLOPE OF 1/4" PER FOOT. INDIVIDUAL UNIT CONDENSATE DRAINS SHALL BE 3/4" DIAMETER LINES AND 1" FOR COMBINED LINES UNLESS OTHERWISE INDICATED.
- THE CONTRACTOR SHALL PROVIDE ALL CONTROL COMPONENTS AND ACCESSORIES INCLUDING EQUIPMENT MOTOR STARTERS, THERMOSTATS, SENSORS, WIRING, BOXES, ETC.
- INSULATE ALL SEWER PIPING RECEIVING AIR CONDITIONING CONDENSATE DRAINS, OR ANY OTHER COLD LIQUID WHICH MAY CREATE CONDENSATION, FROM POINT OF CONNECTION TO TOP OF CONCRETE SLAB-ON-GRADE.
- MOUNT ALL ROOM THERMOSTATS AT 4'-0" ABOVE FINISHED FLOOR UNLESS OTHERWISE INDICATED.
- ALL HVAC SYSTEM THERMOSTATS IN UNITS SHALL BE EQUIPPED WITH ENERGY STAR SEVEN DAY PROGRAMMABLE THERMOSTATS WITH NIGHT SETBACK.
- PRIOR TO PERMIT BEING FINAL, A COMPLETE REPORT OF THE TESTING AND ADJUSTING SHALL BE PROVIDE TO THE OWNER/OWNER'S REPRESENTATIVE AND TO THE INSPECTOR.
- CONDENSER, EVAPORATOR COIL AND FURNACE TO MEET CA.24 REQUIREMENTS.
- INSULATION INSPECTION AND APPROVAL SHALL BE DONE BY APPROVED HERS TESTED(AS REQUIRED).THE INSPECTION SHALL COINCIDE WITH INSULATION INSTALLATION COMPLETION. AND PRIOR TO DRY WALL INSTALLATION(AS REQUIRED BY TITLE 24 ENERGY CALCULATIONS)
- ALL HVAC EQUIPMENT SHALL BE APPROVED PRIOR TO INSTALLATION BY NATIONALLY RECOGNIZED STANDARD AND EVIDENCED BY THE LISTING AND LABEL OF AN APPROVED AGENECY(CMC 2019)
- AIR CONDITIONING EQUIPMENT SHALL NOT BE LOCATED IN SETBACK AREA.
- HEATING AND AIR CONDITIONING SYSTEM DESIGN SHALL BE SIZED, DESIGNED AND HAVE THEIR EQUIPMENT SELECTED USING THE FOLLOWING METHOD(CGBSC 2019).
  A. HEATING LOSS AND HEAT GAIN IS ESTABLISHED ACCORDING TO ACCA MANUAL 3, ASHRAE HANDBOOK OR OTHER EQUIVALENT DESIGN SOFTWARE OR METHOD.
  B. DUCT SYSTEM ARE SIZE ACCORDING TO ACCA 29-D MANUAL D,ASHRAE HANDBOOK OR OTHER EQUIVALENT METHOD.
  C. SELECT HEATING AND COOLING EQUIPMENT ACCORDING TO ACCA 36-S MANUAL S OR EQUIVALENT DESIGN SOFTWARE OR METHOD.
- EQUIPMENT INSTALLED OUTSIDE THE BUILDING SHALL BE LISTED FOR OUTDOOR INSTALLATION OR BE APPROVED WEATHER PROOF ENCLOSURE PER CMC 2019.
- COORDINATE LOCATION OF THE CONDENSATE DRAIN WITH PLUMBING CONTRACTOR .PIPE PRIMARY CONDENSATE TO AN APPROVED RECEPTACLE.
- THE PAD SUPPORTING THE CONDENSER OR COMPRESSOR FROM GROUND SHALL BE NOT LESS THAN 3" ABOVE GRADE CMC 2019.
- AIR CONDITIONING EQUIPMENT SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURERS INSTRUCTION.UNLESS THE EQUIPMENT IS LISTED FOR INSTALLATION ON A COMBUSTIBLE SURFACE SUCH AS A FLOOR OR ROOF OR UNLESS THE SURFACE IS PROTECTED IN APPROVED MANNER,IT SHALL BE INSTALLED ON A SURFACE OF NONCOMBUSTIBLE CONSTRUCTION WITH NO COMBUSTIBLE MATERIAL AGAINST THE UNDERSIDE THEROF(CMC 2019).
- PROVIDE REFRIGERANT SERVICE PORT WITH REQUIRE LOCKING TYPE,TAMPER PROOF REFRIGERANT CAP(CMC 2019).
- CONDENSATE FROM AIR WASHER,AIR COOLING COIL,OR SIMILAR AIR-CONDITIONING EQUIPMENT SHALL BE COLLECTED AND DISCHARGED TO AN APPROVED PLUMBING FIXTURE OR DISPOSAL AREA. (CPC 2019).
- FURNACE MUST SEALED TO PLATFORMS TO PREVENT AIR LEAKAGE AND AIR INFILTRATION.
- COMBUSTION AIR TO FURNACE CLOSETS SHALL COMPLY 2019 CPC AND 2019 CMC.
- THE FURNACE COMPARTMENT SHALL PROVIDE 3" MINIMUM CLEARANCE ON SIDE AND BACK AND 6" FROM THE FRONT OF EQUIPMENT TO THE COMBUSTION AIR INTAKE PER CMC 2019.
- COMBUSTION AIR INTO FURNACE ENCLOSURE SHALL HAVE TWO OPENINGS,EACH HAVING ONE SQ.INCH PER 1000 BTUIH INPUT.(100 SQ.IN.MIN.) OPENING SHALL BE COVERED WITH CORROSION-RESISTANT SCREEN OF 1/4" MESH PER CMC 2019.
- THE FAU CLOSET MUST BE 12" WIDER THAN THE FURNACE OR FURNACES BEING INSTALLED. (CMC 2019)
- PROVIDE SMOOTH METAL DUCT FOR DRYER EXHAUST NOT LESS THAN 4" AND SHALL TERMINATE OUTSIDE THE BUILDING PER CMC 2019.
- DUCT USED FOR KITCHEN RANGE VENTILATION SHALL BE METAL AND SHALL HAVE SMOOTH INTERIOR SURFACE. (CMC 2019)
- DUCT PENETRATION IN THE GARAGE AND DUCT PENETRATING THE WALLS OR CEILINGS SEPARATING THE DWELLING ROOM FROM THE GARAGE SHALL BE CONSTRUCTED OF A MINIMUM NO.26 GAUGE SHEET STEEL OR OTHER APPROVED MATERIAL SHALL HAVE NO OPENINGS INTO THE GARAGE.
- FASTENERS SHALL BE UL 181 APPROVED AND BE INSTALLED WITHOUT 24 ENERGY CALCULATIONS.USE MINIMUM 2' WIDE HANGER MATERIAL.(602.4).
- DUCT LINER SHALL BE FASTENED WITH BOTH APPROVED ADHESIVE AND MECHANICAL FASTENERS,EXPOSED EDGES SHALL BE TREATED AS NOT TO CAUSE DETERIORATION OF THE LINER(605.4).
- EXTERNAL DUCT INSULATION SHALL BE LABELED WITH MANUFACTURER' NAME, THE R VALUE, THE FLAME SPREAD INDEX AND THE SMOKE DEVELOPMENT INDEX(604.4)
- DUCTWORK SHALL BE CONSTRUCTED IN ACCORDANCE WITH CHAPTER 6 OF THE CMC OR APPLICABLE SMACNA STANDARDS.
- DUCT JOINTS SHALL BE SEALED WITH MATERIAL TO MEET TITLE 24 REQUIREMENTS AND CURRENT BUILDING CODES.
- THERMOSTATS WILL COMPLY WITH CAL.TITLE 24 REGULATIONS.
- THERMOSTAT AND WIRING WHICH WILL CONTROL BOTH HEATING AND AIR CONDITIONING EQUIPMENTS.EACH HVAC SYSTEM SHALL BE APPROVED WITH AT LEAST ONE AUTOMATIC TEMPERATURE CONTROL DEVICE FOR REGULATION OF TEMPERATURE AS REQUIRED IN SECTION T-1503(C) OF THE CALIFORNIA ADMINISTRATIVE CODE.
- ALL THERMOSTAT SHALL BE ELECTRONIC PROGRAMMABLE TYPE 7 DAY,INSTALL LOW VOLTAGE WIRING AS REQUIRED
- MECHANICAL EXHAUST FANS WHICH EXHAUST DIRECTLY FROM BATHROOMS SHALL BE ENERGY STAR,DUCTED TO TERMINATE OUTSIDE THE BUILDING AND UNLESS FUNCTIONING AS A COMPONENT OF A WHOLE HOUSE VENTILATION SYSTEM BE CONTROLLED BE A READILY ACCESSIBLE HUMIDISTAT SHALL MAINTAIN HUMIDITY BETWEEN 50-80% RH(CGBSC 2019).
- A VENTILATION FAN SHALL BE FURNISHED PER TITLE 24 ENERGY CODE TO RUN CONTINUOUS.THE CAPACITY OF THE FAN SHALL BE CALCULATED PER TITLE 24.A SIGN SHOULD STATE THAT THE FAN SHOULD RUN CONTINUOUSLY FOR PROPER RESIDENCE VENTILATION.
- NOT MORE THAN ONE APPLIANCE SHALL BE CONNECTED TO SINGLE VENT UNLESS CALCULATION SHOWING CONFORMANCE TO CMC 2019 CHAPTER & SUBMITTED TO BUILDING AND SAFETY DEPARTMENT.
- PROVIDE MIN.30" CLEARANCE ABOVE THE RANGE OR COOK TOP TO UNPROTECTED COMBUSTIBLE MATERIAL OR 24" CLEARANCE TO METAL VENTILATION HOOD(CMC 2019).

HVAC SHEET INDEX

SR NO.	SHEET NO.	SHEET NAME	SCALE
1	HG.1	GENERAL NOTES AND SYMBOL	N.T.S.
2	H1.0	MECHANICAL FLOOR AND ROOF PLAN	1/4" = 1'-0"
3	H2.0	MECHANICAL PIPING PLAN	1/4" = 1'-0"
4	H3.0	GAS RISER DIAGRAM	N.T.S.
5	H4.0	HVAC DETAILS	N.T.S.
6	H5.0	HVAC SCHEDULE	N.T.S.

REFERENCE CODES
1) CALIFORNIA BUILDING CODE, 2019 EDITION. 2) CALIFORNIA RESIDENTIAL CODE, 2019 EDITION. 3) CALIFORNIA MECHANICAL CODE, 2019 EDITION. 4) CALIFORNIA PLUMBING CODE, 2019 EDITION. 5) CALIFORNIA ELECTRICAL CODE, 2019 EDITION.

EQUIPMENT IDENTIFICATION

XXX — SYMBOL OR TYPE (SEE SCHEDULE OR SPECIFICATIONS)
X — IDENTIFICATION NUMBER (SEE SCHEDULE)

XX — KEYNOTES

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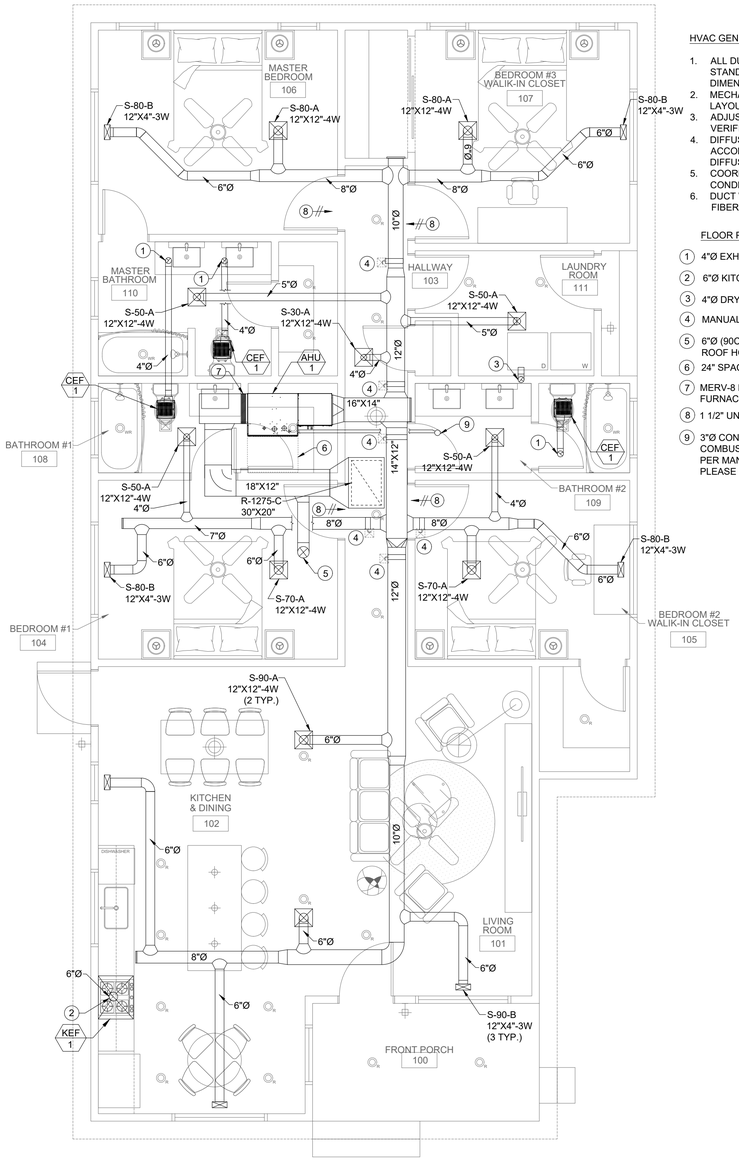
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REV. DATE.	REV.

GENERAL NOTES AND SYMBOLS

HG.1



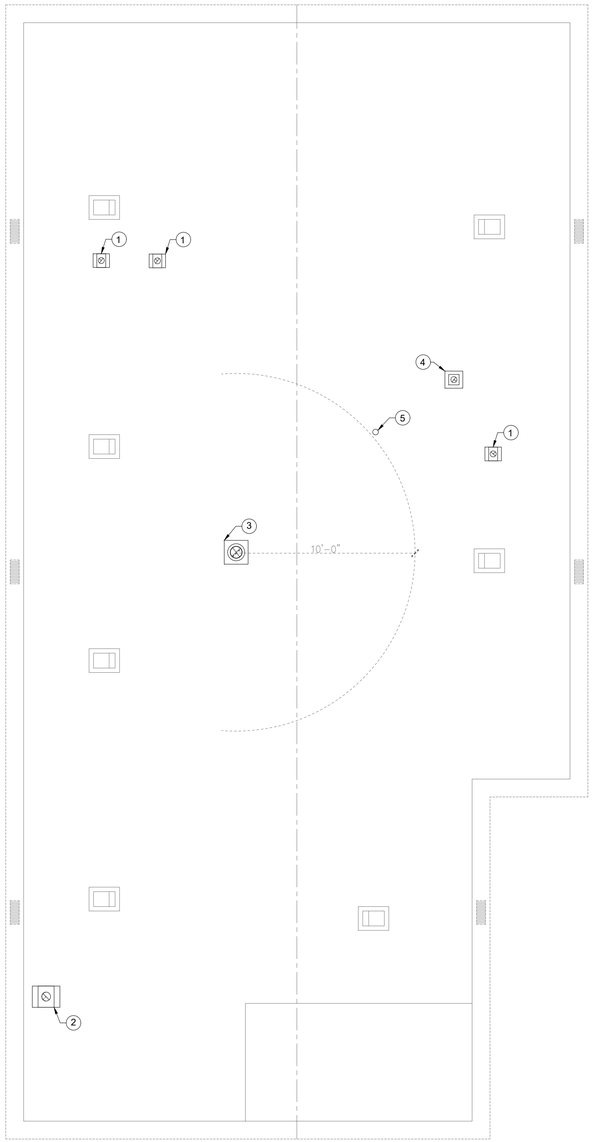
N 1 MECHANICAL FLOOR PLAN
SCALE: 1/4" = 1'-0"

HVAC GENERAL NOTES

1. ALL DUCTWORK SHALL CONFORM TO SMACNA STANDARDS.ALL DUCTWORK SIZES ARE NET INSIDE DIMENSIONS.
2. MECHANICAL CONTRACTOR TO COORDINATE DUCT LAYOUT AND DIMENSIONS WITH FIELD VERIFIED.
3. ADJUST CEILING DIFFUSER TO FIT AS PER FIELD VERIFIED.
4. DIFFUSER TO BE INSTALLED WITH SYMMETRY IN ACCORDANCE WITH LIGHTING LAYOUT DESIGN. DIFFUSER SHALL NOT BE OFF CENTER OR ALIGNMENT.
5. COORDINATE WITH GENERAL CONTRACTOR FOR ALL CONDENSATE DRAIN LINE LOCATION.
6. DUCT WORK SHALL BE INSULATED WITH R-8 FIBERGLASS DUCT WRAP WITH FSK JACKET.

FLOOR PLAN KEYNOTES:

1. 4"Ø EXHAUST DUCT UP TO ROOF HOOD.
2. 6"Ø KITCHEN HOOD EXHAUST DUCT UP TO ROOF HOOD.
3. 4"Ø DRYER DUCT EXHAUST THROUGH ROOF HOOD.
4. MANUAL AIR DAMPER WITH ACCESS PANEL.
5. 6"Ø (90CFM) OUTSIDE AIR DUCT THROUGH ROOF HOOD.
6. 24" SPACE REQUIRED FOR AHU SERVICE.
7. MERV-8 FILTER KIT CONNECTED WITH FURNACE RETURN.
8. 1 1/2" UNDERCUT DOOR.
9. 3"Ø CONCENTRIC VENT KIT. COMBINATION FLUE AND COMBUSTION AIR THROUGH ROOF. INSTALLATION AS PER MANUFACTURER'S INSTRUCTION. PLEASE REFER SHEET:M4.0, DETAIL:4.



N 2 MECHANICAL ROOF PLAN
SCALE: 1/4" = 1'-0"

HVAC GENERAL NOTES

1. ALL DUCTWORK SHALL CONFORM TO SMACNA STANDARDS.ALL DUCTWORK SIZES ARE NET INSIDE DIMENSIONS.
2. MECHANICAL CONTRACTOR TO COORDINATE DUCT LAYOUT AND DIMENSIONS WITH FIELD VERIFIED.
3. ADJUST CEILING DIFFUSER TO FIT AS PER FIELD VERIFIED.
4. DIFFUSER TO BE INSTALLED WITH SYMMETRY IN ACCORDANCE WITH LIGHTING LAYOUT DESIGN. DIFFUSER SHALL NOT BE OFF CENTER OR ALIGNMENT.
5. COORDINATE WITH GENERAL CONTRACTOR FOR ALL CONDENSATE DRAIN LINE LOCATION.
6. DUCT WORK SHALL BE INSULATED WITH R-8 FIBERGLASS DUCT WRAP WITH FSK JACKET.

ROOF KEYNOTES:

1. 4"Ø EXHAUST DUCT THROUGH ROOF HOOD WITH BUILT IN BACK DRAFT DAMPER & BIRD SCREEN. (MANUFACTURER #BROAN, MODEL NO:636)
2. 6"Ø EXHAUST DUCT EXHAUST THROUGH ROOF HOOD WITH BUILT-IN BACK DRAFT DAMPER & BIRD SCREEN. (MANUFACTURER #BROAN MODEL NO:634M OR EQUIVALENT)
3. 6"Ø (90 CFM) AIR INTAKE DUCT THROUGH ROOF HOOD. (MANUFACTURER #BROAN, MODEL:611CM)
4. 4"Ø DRYER EXHAUST THROUGH ROOF HOOD. (MANUFACTURER #DRYERJACK, MODEL #477)
5. 3"Ø CONCENTRIC VENT KIT. COMBINATION FLUE AND COMBUSTION AIR THROUGH ROOF. INSTALLATION AS PER MANUFACTURER'S INSTRUCTION. PLEASE REFER SHEET:M4.0, DETAIL:4.

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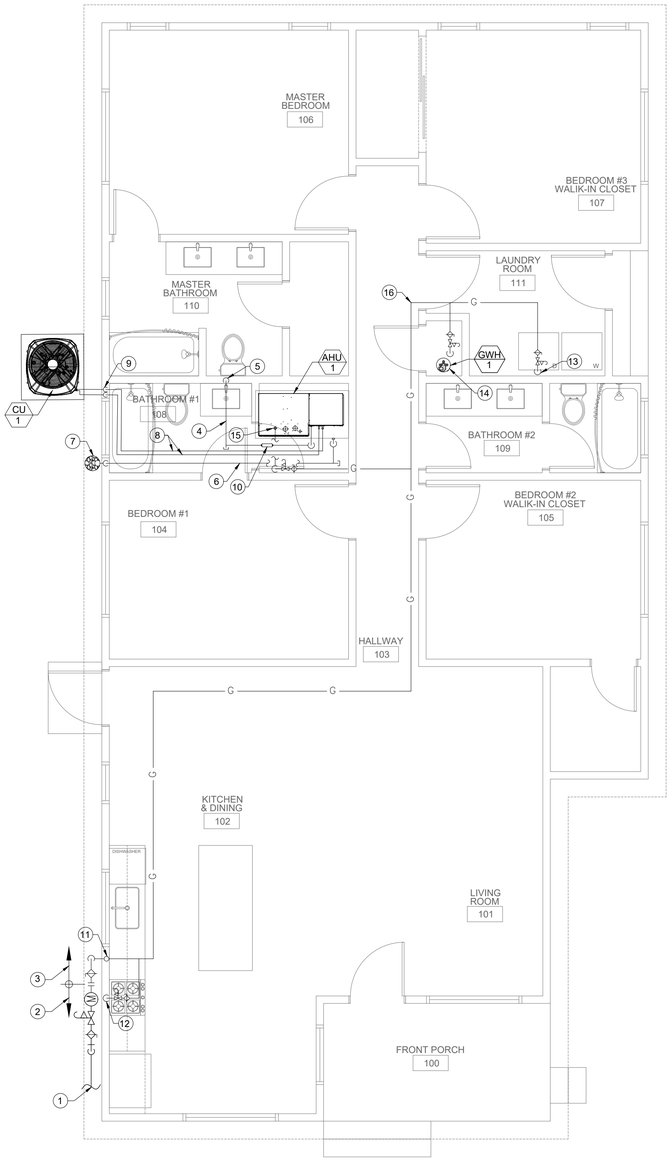
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HVAC FLOOR AND ROOF PLAN

H1.0



KEYNOTES:

1. INCOMING GAS SERVICE BY GAS UTILITY COMPANY.
2. BY GAS UTILITY COMPANY.
3. BY PLUMBING CONTRACTOR.
4. 1"Ø INSULATION CONDENSATE DRAIN PIPE. (PIPE MATERIAL PVC)
5. CONDENSATE DRAIN PIPE CONNECT TO LAVATORY DRAIN. REFER SHEET NO:H4.0, DETAIL:13.
6. 1"Ø SECONDARY DRAIN PIPING.
7. 1"Ø SECONDARY DRAIN PIPE TERMINATE AT DRY WELL WITH INDIRECT CONNECTION.
8. REFRIGERATION PIPING SHALL BE COPPER ASTM B-280 ACR TUBING. WITH INSULATION.
9. REFRIGERATION PIPING DOWN.
10. NEUTRALIZATION KIT.
11. 1 1/4"Ø (195 CFH) GAS PIPE COMING UP.
12. 1/2"Ø (60 CFH) GAS PIPE TO BURNER.
13. 1/2"Ø (35 CFH) GAS PIPE TO DRYER.
14. 1/2"Ø (60 CFH) GAS PIPE TO WATER HEATER (WH-1).
15. 1/2"Ø (40 CFH) GAS PIPE TO AHU-1.
16. 3/4"Ø (95 CFH) GAS PIPE.
17. 1"Ø (135 CFH) GAS PIPE.
18. 1 1/4"Ø (195 CFH) GAS PIPE.

N

1 MECHANICAL PIPING FLOOR PLAN
SCALE: 1/4" = 1'-0"

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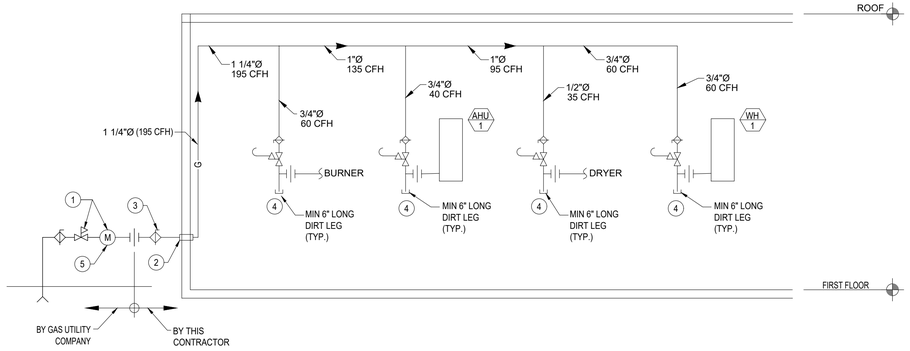
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HVAC AND GAS PIPING FLOOR PLAN

H2.0



1 GAS RISER DIAGRAM PLAN
SCALE: N.T.S.

NATURAL GAS CONSUMPTION SCHEDULE SCHEDULE 40 METALLIC PIPE FROM UPC CODE INLET PRESSURE : LESS THAN 2PSI. PRESSURE DROP :0.5 INCH WC.							
DESCRIPTION	MANUFACTURER	MODEL	REMARKS	QTY	CFH	TOTAL CFH	DISTANCE FROM CONNECTION (FT)
GAS FURNACE (AHU-1)	TRANE	S9X1B040U3PSBA	NEW	1	40	40	70
GAS WATER HEATER (WH-1)	RHEEM	PRO+G29-60N RH70 ID	NEW	1	60	60	70
DRYER (GROUND FLOOR)	-	-	NEW	1	35	35	80
KITCHEN RANGE (BURNER)	-	-	NEW	1	60	60	25
					TOTAL	195	

DRAWING NOTES:

1. INCOMING GAS SERVICE, PRESSURE REGULATOR, AND METER BY GAS UTILITY COMPANY TO DELIVER INDICATED GAS FLOW RATE @ 2.0 PSI DOWNSTREAM OF PRESSURE REGULATOR AND METER.
2. ENTER BUILDING APPROXIMATELY 2'-6" TO 3'-0" ABOVE FINISHED FLOOR (V.I.F.) WITH PIPE SLEEVE THROUGH EXTERIOR WALL.
3. SHUT-OFF VALVE FOR "BUILDING NATURAL GAS MAIN SHUT-OFF VALVE".
4. MAKE FULL SIZE CONNECTION TO GAS FIRED EQUIPMENT WITH MINIMUM 6" LONG DIRT LEG.
5. METER IS LOCATED IN THE CLUSTER OF METERS IN THE NORTH-EAST SIDE OF BUILDING FIELD VERIFY.

GAS PIPING NOTES

1. 2 INCHES AND UNDER: SCHEDULE 40, BLACK, ASTM A53, GRADE A, SEAMLESS OR LAP WELDED. FITTINGS: STANDARD WEIGHT, BLACK STEEL THREADED WITH MALLEABLE FITTINGS.
2. OVER 2 INCHES: SCHEDULE 40, BLACK, ASTM A53, GRADE A, SEAMLESS OR LAP WELDED. FITTINGS: STANDARD WEIGHT, BLACK STEEL BUTT WELDING. JOINTS: FLANGED OR WELDED, RAISED FACE 150 LB.
3. ALL GAS PIPING AND FITTINGS SHALL BEAR THE MANUFACTURERS' IDENTIFICATION AND BE LISTED BY A THIRD PARTY CERTIFICATION AGENCY.
4. OUTSIDE PIPING, FITTINGS, AND VALVES SHALL HAVE ONE COAT PRIMER AND ONE FINISH COAT YELLOW.
5. GAS COCKS TO BE SUITABLE FOR GAS SERVICE.
6. INTERIOR GAS PIPING SHALL NOT RUN IN OR THROUGH AN AIR DUCT, CLOTHES CHUTE, CHIMNEY OR VENT, VENTILATING DUCT, DUMBWAITER, ELEVATOR SHAFT, OR CONCEALED PLENUM.
7. PIPING SHALL NOT BE LOCATED IN SOLID PARTITIONS OR SOLID WALLS, UNLESS INSTALLED IN A CHASE OR CASING.
8. PIPING INSTALLED IN CONCEALED LOCATIONS SHALL BE WELDED.
9. PIPING SHALL NOT BE USED AS A GROUNDING ELECTRODE.
10. PIPING INSTALLED THROUGH A MASONRY WALL SHALL BE ENCASED IN A SLEEVE. THE SLEEVE SHALL BE SEALED AT THE OUTSIDE OF THE FOUNDATION WALL TO PREVENT ENTRY OF WATER. GAS PIPING SHALL NOT PENETRATE A BUILDING FOUNDATION WALL BELOW GRADE.
11. PROVIDE PIPE SUPPORT FOR ROOFTOP GAS PIPING. PIPE SUPPORT SHALL BE AS MANUFACTURED BY MIRO INDUSTRIES. MAINTAIN MAXIMUM LOAD WEIGHT AND SPACING AS RECOMMENDED BY MANUFACTURER.
12. GAS PRESSURE REGULATORS THAT REQUIRE A VENT SHALL HAVE AN INDEPENDENT VENT TO THE OUTSIDE OF THE BUILDING. THE VENT SHALL BE TERMINATED SO AS TO PREVENT THE ENTRY OF WATER OR FOREIGN OBJECTS.
13. SCHEDULE 40 METALLIC GAS PIPE SIZED FOR CONSIDERING 2 PSI AND 0.5 IN W.C PRESSURE DROP.

GAS PRESSURE REGULATOR VENT NOTE

GAS PRESSURE REGULATORS INSTALLED INDOORS SHALL HAVE FULL SIZED VENT PIPED TO OUTDOORS AND TERMINATE WITH SCREENED ELBOW TURNED DOWN A MINIMUM OF 12" ABOVE ROOF OR GRADE.

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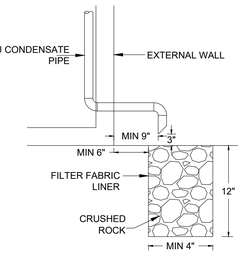
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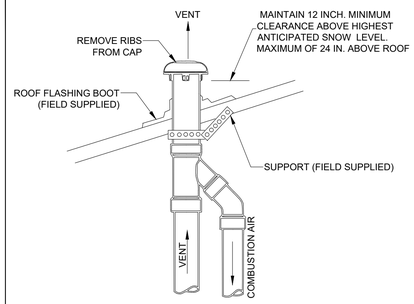
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GAS RISER DIAGRAM

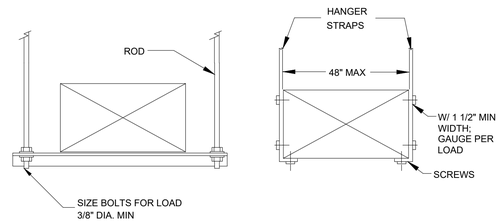
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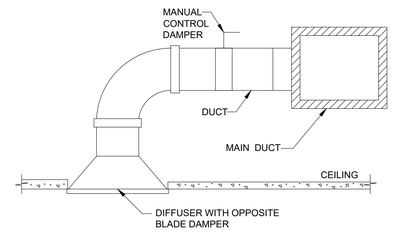
DRY WELL DETAIL — 5
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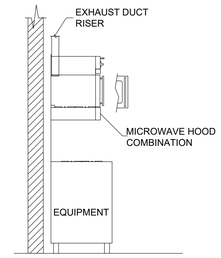
FURNACE CONCENTRIC VENT DETAIL — 4
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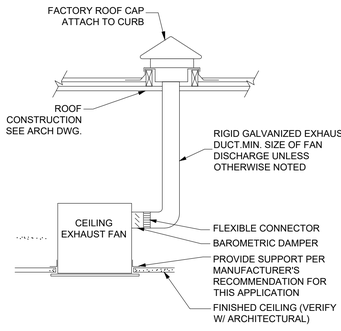
HANGER SUPPORT DETAIL — 3
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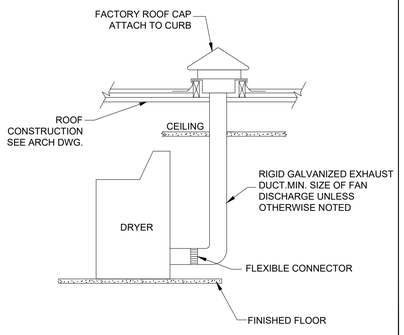


CEILING DIFFUSER BRANCH DUCT DETAIL — 2
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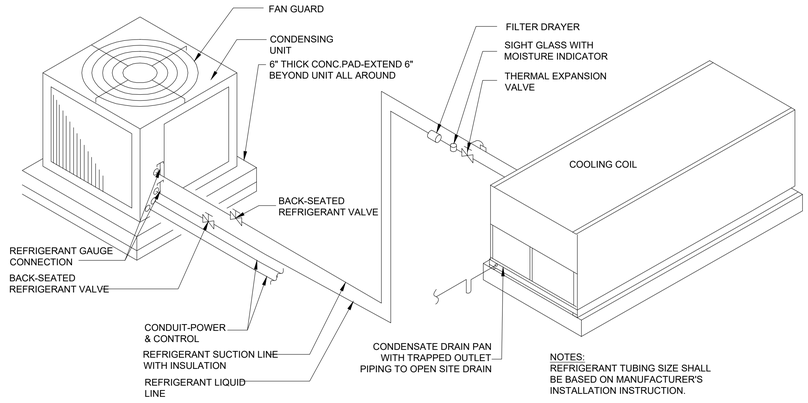


KITCHEN VENT HOOD/MICROWAVE COMBO DETAIL — 1
NOT TO SCALE

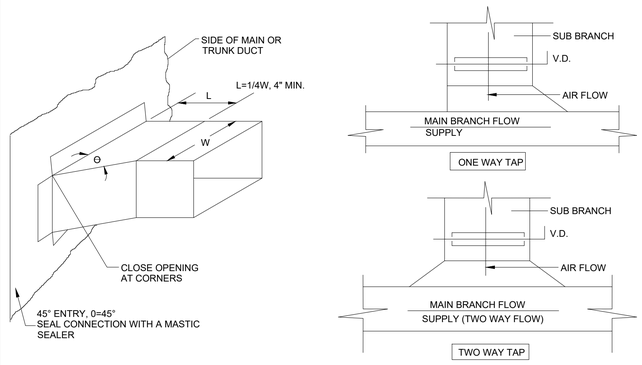




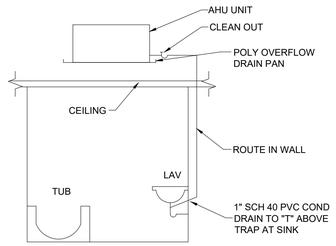
TOILET, LAUNDRY AND DRYER ROOF EXHAUST DETAIL — 8
NOT TO SCALE



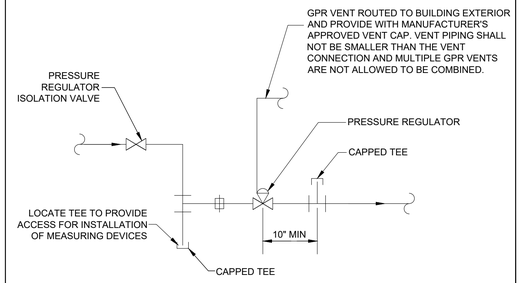
CONDENSING UNIT DETAIL — 7
NOT TO SCALE



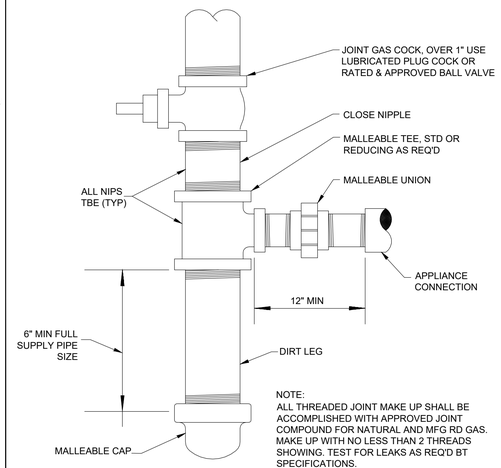
TYPICAL BRANCH CONNECTION DETAILS — 6
NOT TO SCALE



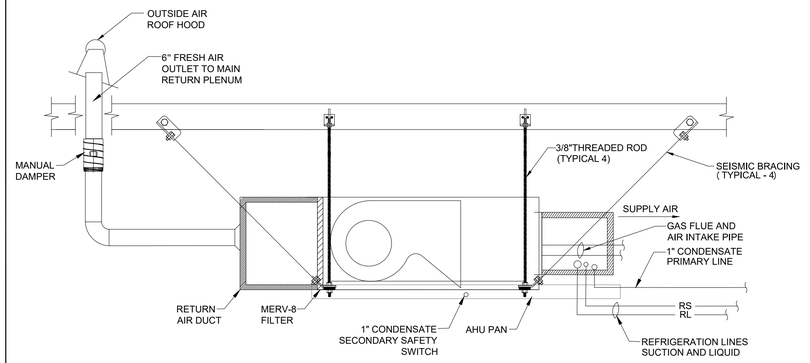
CONDENSATE TAP DETAIL — 13
NOT TO SCALE



GAS PRESSURE REGULATING DETAIL — 11
NOT TO SCALE



GAS CONNECTION DETAIL — 10
NOT TO SCALE



FURNACE INSTALLATION DETAIL — 9
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DETAILS
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GAS FURNACE SCHEDULE

SYMBOL	TYPE	LOCATION	QTY.	SUPPLY CFM	E.S.P. (IN. W.C.)	INPUT HEATING CAPACITY (MBTU)	HEATING AND COOLING STAGE	FLUE OUTLET SIZE (INCH.)	BLOWER DRIVE SECTION				GAS			ELECTRICAL					SHIPPING WEIGHT (LBS)	DIMENSION (IN INCH.)			AFUE (%)	MODEL	BASIS OF DESIGN
									FAN		MOTOR																	
									QTY.	DIAMETER	FLA	HP	V./PH./HZ	BURNERS TYPE	BURNERS QUANTITY	PIPE SIZE (INCH.)	AMPACITY (IN AMPS)	MOP D	V.	HZ	PH.		WIDTH	LENGTH	HEIGHT			
AHU-1	HORIZONTAL TOP FLOW	IN ATTICS	1	1350	0.5	40	2	2"ø	1	11"x8"	6.4	1/2	120/1/60	MULTIPORT INSHOT	2	1 1/2"	7.9	15	120	1	60	122	17 1/2"	28 3/4"	34"	96	S9V2B040U3PSAA	TRANE

NOTES:
1. ALL FLUE GAS CARRYING AREAS EXTERNAL TO THE FURNACE (I.E. CHIMNEY, VENT CONNECTOR) ARE CLEAR AND FREE OF OBSTRUCTIONS. AND MAXIMUM TOTAL EQUIVALENT LENGTH OF FLUE PIPE IS 60 FEET.
2. THE VENT CONNECTOR IS IN PLACE, SLOPES UPWARD AND IS PHYSICALLY SOUND WITHOUT HOLES OR GAPS.
3. THE RETURN AIR DUCT CONNECTION IS PHYSICALLY SOUND, IS SEALED TO THE FURNACE CASING, AND TERMINATES OUTSIDE THE SPACE CONTAINING THE FURNACE.
4. THE BURNER FLAMES ARE IN GOOD ADJUSTMENT.
5. PROPYLENE GLYCOL NEED NOT BE REMOVED BEFORE RE-STARTING FURNACE.
6. UNIT WITH RETURN AIR FILTER KIT.

OUTDOOR UNIT SCHEDULE

SYMBOL	LOCATION	QTY.	COOLING RATED CAPACITY (MBH)	ENTERING AIR DB (°F)	REFRIGERANT		COMPRESSOR				OUTDOOR FAN		ELECTRIC					SHIPPING WEIGHT (LBS)	DIMENSION (IN INCH.)			TYPE OF REFRIGERANT	SEER	MODEL	BASIS OF DESIGN
					SUCTION (INCH.)	LIQUID (INCH.)	TYPE	STAGE	QTY.	RLA. (AMPS)	FAN QTY.	MOTOR HP	MCA	MOCP	V.	PH.	HZ.		WIDTH	LENGTH	HEIGHT				
CU-1	OUTSIDE BUILDING	1	36	105	3/4"	3/8"	CLIMATUFF _SCROLL	2	1	17	1	1/8	24	35	240	1	60	316	35 1/4"	38 3/4"	50"	R-410A	18	4TTX8036A1000A	TRANE

NOTES:
1. OUTDOOR UNITS SHALL BE MOUNTED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTION.
2. INSTALL THE AIR CONDITIONER ON A STRONG ENOUGH TO WITHSTAND THE WEIGHT OF THE UNIT.
3. ELECTRICAL WORK MUST BE PERFORMED IN ACCORDANCE WITH RELEVANT LOCAL AND NATIONAL REGULATIONS AND WITH INSTRUCTIONS IN THIS INSTALLATION MANUAL.
4. DURING INSTALLATION, ATTACH THE REFRIGERANT PIPING SECURELY BEFORE RUINNING THE COMPRESSOR.
5. MAX. LENGTH OF REFRIGERANT LINES FROM OUTDOOR TO INDOOR UNIT MUST NOT EXCEED 60 FEET. THE VERTICAL CHANGES MUST NOT EXCEED 25 FEET.

COOLING COIL(INDOOR UNIT) SCHEDULE

SYMBOL	LOCATION	QTY.	TOTAL COOLING CAPACITY (MBH)	DIMENSION (INCH.)			TYPE OF REFRIGE RANT	SHIPPING WEIGHT (LBS.)	MODEL	BASIS OF DESIGN
				LENGTH	WIDTH	HEIGHT				
CC-1	IN ATTICS	1	36	21 1/2"	17 1/2"	22 1/2"	R-410A	54	4PXCBU36BS3HAA	TRANE

NOTES:
1. COOLING COIL SHALL BE MOUNTED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTION.
2. REFRIGERANT PIPING SHALL RUN FROM INDOOR UNITS TO OUTDOOR UNIT AS PER MANUFACTURER'S INSTRUCTION AND BOTH REFRIGERANT PIPING SHALL BE INSULATED.

CEILING EXHAUST FAN SCHEDULE

SYMBOL	TYPE	LOCATION	QTY.	FAN					PHYSICAL DATA				SOUND (SONES)	MODEL	BASIS OF DESIGN
				AIR FLOW	E.S.P.	ELECTRICAL			DIMENSION (INCH.)			NET WEIGHT (LB)			
				CFM	IN H2O	V.	HZ	WATTS	LENGTH	WIDTH	HEIGHT				
CEF-1	CEILING MOUNTED	TOILET	3	50	0.1	120	60	20	9 1/4"	9 1/4"	3 5/8"	10	0.5	AE50	BROAN

NOTES:
1. TO AVOID MOTOR BEARING DAMAGE AND NOISY OR UNBALANCED IMPELLERS, KEEP DRYWALL SPRAY, CONSTRUCTION DUST, ETC. OFF POWER UNIT.
2. IT SHALL BE DUCTED TO A ROOF OR WALL CAP USING ROUND DUCTWORK.
3. BLOWER ASSEMBLY SHALL BE REMOVABLE, HAVE A CENTRIFUGAL TYPE BLOWER WHEEL AND A PERMANENTLY LUBRICATED.
4. MOTOR SHALL BE DESIGN FOR CONTINUOUS OPERATION.

RESIDENTIAL MICROWAVE RECIRCULATION HOOD SCHEDULE

SYMBOL	TYPE	LOCATION	QTY.	CAPACITY (CU.FT.)	VENTILATION SYSTEM					ELECTRICAL				TYPE OF LIGHTS	MICRO WAVE DIMENSION(INCH.)			MODEL	BASIS OF DESIGN
					BLOWER TYPE	CFM	NUMBER OF SPEED	VENTILATION TYPE	FILTER TYPE	V.	PH.	HZ.	POWER (WATTS)		WIDTH	DEPTH	HEIGHT		
KEF-1	RECIRCULATION KITCHEN HOOD/MICROWAVE COMBO	KITCHEN	1	1.10	CENTRIFUGAL	250	4	UPDRAFT	CHARCOAL ODOR REMOVING FILTER	120	1	60	1000	LED	19"	15"	7"	WML75011HV	WHIRLPOOL

NOTES:
1. INSTALL ACCORDANCE WITH MANUFACTURER'S INSTRUCTION.

AIR DEVICE SCHEDULE

				NOMENCLATURE	U = USAGE (S = SUPPLY, R = RETURN, E = EXHAUST, T = TRANSFER			EXAMPLE	REFERENCE IMAGE
				U-CFM-T	CFM = AIR QUANITY		T = TYPE OF AIR DEVICE	S-150-A	
				SIZE- B	SIZE = NECK SIZE OF AIR DEVICE		B = BLOW PATTERN	8"Ø-4W	
TYPE	USAGE	SERVING	NECK SIZE	PANEL/FACE SIZE	MATERIAL	DESCRIPTION	MANUFACTURER	MODEL NO.	REMARKS	
A	SUPPLY	REFER TO FLOOR PLAN	SEE PLAN	12"x12"	ALUMINUM	4-WAY AIR DIFFUSER WITH OPPOSED BLADE DAMPER	HART & COOLEY	A504-OB	1,2,3,4,5,6	
B	SUPPLY	REFER TO FLOOR PLAN	12"X4"	13 1/2"X5 1/2"	ALUMINUM	ALUMINUM 3-WAY SUPPLY AIR REGISTER WITH MULTI SHUTTER	HART & COOLEY	A683	1,2,3,4,5,6	
C	RETURN	REFER TO FLOOR PLAN	30"X20"	31 3/4"X21 3/4"	ALUMINUM	ALUMINUM RETURN AIR GRILL	HART & COOLEY	A672	1,2,3,4,5,6	

NOTES:
1. CUSTOM FINISH & COLOR AS SELECTED BY ARCHITECT/OWNER.
2. PROVIDE OPPOSED BLADE DAMPER.
3. PROVIDE FOAM GASKET SEAL.
4. COORDINATE FRAME TYPE WITH REFLECTED CEILING PLAN & ARCHITECT.
5. INSULATED PLENUM/BACK PAN.
6. ARCHITECT/OWNER SHALL SELECT DIFFUSER BASE ON DRYWALL CEILING .

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SCHEDULES

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